JACC
VOL. ■, NO. ■, 2026

PUBLISHED BY ELSEVIER

Comparative Prognosis of Chagas and Other Cardiomyopathies

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ABSTRACT

BACKGROUND Chagas disease, caused by *Trypanosoma cruzi* parasites, is a common cause of heart failure (HF) in Latin America and has recently been declared endemic in the United States. The authors compared outcomes in Chagasic HF vs ischemic and other nonischemic etiologies of HF with reduced ejection fraction (HFrEF).

OBJECTIVES The aim of this study was to compare clinical outcomes of Chagasic HFrEF vs ischemic and other non-ischemic etiologies.

METHODS Investigator-reported etiology of HFrEF in the ATMOSPHERE, PARADIGM-HF, and GALACTIC-HF trials was categorized as ischemic, valvular, alcoholic, hypertensive, idiopathic, viral, Chagasic, or "other." Time to the composite of first HF hospitalization or cardiovascular death, its components, all-cause death, and stroke was analyzed using Cox models adjusted for baseline characteristics, patient setting, trial, and other potential confounders.

RESULTS Among 23,647 patients (13,381 ischemic, 4,344 idiopathic, 2,559 hypertensive, 1,923 others, 423 alcoholic, 412 valvular, 297 viral, and 308 Chagasic), Chagasic HF had the highest incidence rates of all clinical outcomes compared with other etiologies. Compared with patients with ischemic etiology, the adjusted HRs in Chagasic HF were significantly higher for the composite outcome (HR: 1.65; 95% CI: 1.36-2.02), HF hospitalization (HR: 1.75; 95% CI: 1.36-2.25), cardiovascular death (HR: 1.86; 95% CI: 1.47-2.35), all-cause death (HR: 1.82; 95% CI: 1.47-2.25), and stroke (HR: 2.16; 95% CI: 1.20-3.88).

CONCLUSIONS Patients with Chagasic HFrEF have a distinct clinical course associated not only with excess mortality but also with an increased risk for stroke compared with other etiologies except for valvular and "other" etiologies. (Aliskiren Trial to Minimize Outcomes in Patients with Heart Failure [ATMOSPHERE], NCT00853658; Prospective Comparison of ARNI [Angiotensin Receptor-Neprilysin Inhibitor] with ACEI [Angiotensin-Converting-Enzyme Inhibitor] to Determine Impact on Global Mortality and Morbidity in Heart Failure Trial [PARADIGM-HF], NCT01035255; Global Approach to Lowering Adverse Cardiac Outcomes Through Improving Contractility in Heart Failure [GALACTIC-HF], NCT02929329) (JACC. 2026;■:■-■) © 2026 by the American College of Cardiology Foundation.

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Manuscript received February 9, 2026; revised manuscript received March 2, 2026, accepted March 3, 2026.

ISSN 0735-1097/$36.00
https://doi.org/10.1016/j.jacc.2026.03.022

ABBREVIATIONS AND ACRONYMS

BNP = B-type natriuretic peptide

CCC = chronic Chagas cardiomyopathy

HF = heart failure

HFrEF = heart failure with reduced ejection fraction

LVEF = left ventricular ejection fraction

NT-proBNP = N-terminal pro-B-type natriuretic peptide

Chagas disease, caused by *Trypanosoma cruzi* infection, remains a major yet under-recognized cause of heart failure (HF) across the Americas.[1,2] Once confined to Latin America, Chagas disease has become a growing global concern because of migration, expanding vector distribution, and increasingly underdiagnosed local transmission.[3-6] Current estimates suggest that approximately 6 million to 7 million people worldwide are infected, including approximately 300,000 individuals in the United States, where Chagas disease has recently been declared endemic by the Centers for Disease Control and Prevention.[3,7-9] Many of these individuals are at risk for developing chronic Chagas cardiomyopathy (CCC), making Chagas disease a significant cause of HF even in high-income countries.[3,10] CCC has unique pathophysiological and clinical characteristics, including diffuse myocardial fibrosis, conduction abnormalities, and apical aneurysm formation, with a strong predisposition to arrhythmias and thromboembolic events, including stroke.[11-13] We previously reported that Chagasic HF is associated with substantially higher morbidity and mortality compared with ischemic and nonischemic HF.[12] However, cerebrovascular events were not assessed, nonischemic etiologies were grouped together as a single category, and the study population was limited to Latin America. To better understand how the prognosis of Chagasic HF compares with other nonischemic etiologies—such as valvular, hypertensive, alcoholic, and idiopathic cardiomyopathies—we performed an individual participant-level analysis combining data from 3 global trials of HF with reduced ejection fraction (HFrEF) that collected information on nonischemic etiology, including Chagas disease. We compared clinical characteristics and outcomes, including mode of death and cerebrovascular events, between Chagasic HF and other nonischemic etiologies, extending the evaluation beyond Latin America.

METHODS

Individual participant-level data from 3 trials of HFrEF (ATMOSPHERE [Aliskiren Trial to Minimize Outcomes in Patients With Heart Failure], PARADIGM-HF [Prospective Comparison of ARNI With ACEI to Determine Impact on Global Mortality and Morbidity in Heart Failure], and GALACTIC-HF [Global Approach to Lowering Adverse Cardiac Outcomes Through Improving Contractility in Heart Failure]) were included in the present study. The design, baseline characteristics, and results of these trials have been published in detail previously.[14-16] A summary of the designs is provided in Supplemental Table 1. Briefly, in ATMOSPHERE, patients had NYHA functional class II to IV symptoms, HF with reduced left ventricular ejection fraction (LVEF; ≤35%), and elevated plasma natriuretic peptide levels (B-type natriuretic peptide [BNP] ≥150 pg/mL or N-terminal pro-BNP [NT-proBNP] ≥600 pg/mL). If a patient had been hospitalized for HF within 12 months, the eligibility criterion for plasma natriuretic peptide level was BNP ≥100 pg/mL or NT-proBNP ≥400 pg/mL. In PARADIGM-HF, patients had NYHA functional class II to IV symptoms, LVEFs ≤40% (changed to ≤35% by amendment), and the same elevated plasma natriuretic peptide criteria as in ATMOSPHERE. In GALACTIC-HF, patients had NYHA functional class II to IV symptoms and LVEFs ≤35%. Patients were either hospitalized for worsening HF at screening or had experienced HF-related hospitalizations or emergency department visits within the prior year. All participants were required to be on guideline-directed medical therapy and to have elevated natriuretic peptide levels (NT-proBNP ≥400 pg/mL or ≥1,200 pg/mL in atrial fibrillation, BNP ≥125 pg/mL or ≥375 pg/mL in atrial fibrillation). Each trial was approved by local ethics committees, and written informed consent was obtained from each patient.

PRIMARY ETIOLOGY OF HF. The primary HF etiology was collected at the screening visit using a similar structured case report form in all trials. Investigator-reported etiology was categorized as ischemic, valvular, alcoholic, hypertensive, idiopathic, viral, Chagasic, or "other." In this study, conditions that were originally reported as "other," as well as peripartum, infectious, diabetic, and drug-induced etiologies, were collectively classified as "other" because the numbers of those patients were small compared with other etiologies.

TRIAL OUTCOMES. The primary outcome of each trial is summarized in Supplemental Table 1. The primary endpoints of the trials were as follows: in ATMOSPHERE and PARADIGM-HF, the primary endpoint was the first occurrence of the composite of cardiovascular death or HF hospitalization. The primary composite outcome in GALACTIC-HF was the first occurrence of the composite of HF events or cardiovascular death. All-cause death was a secondary endpoint in PARADIGM-HF and GALACTIC-HF, but not in ATMOSPHERE, in which it was an exploratory endpoint. In the present study, we used a composite of HF hospitalization or cardiovascular death as the primary endpoint. We also separately analyzed all-cause death, noncardiovascular death, and cardiovascular death, including the 2 major types of cardiovascular death—sudden death and pump failure death—as well as first HF hospitalization, and stroke. Each event and cause of death was adjudicated by a central endpoint committee according to the prespecified criteria in each trial.

STATISTICAL ANALYSIS. Baseline patient characteristics at the time of randomization are reported as median (Q1-Q3) or mean ± SD for continuous variables and as counts and percentages for categorical variables. Baseline characteristics were compared across the etiologic categories using analysis of variance for continuous variables and the chi-square test for categorical variables. The Kansas City Cardiomyopathy Questionnaire clinical summary score and NT-proBNP were not normally distributed and therefore are summarized as median (Q1-Q3) and were compared using the Dunn test for multiple nonparametric pairwise comparisons, with Bonferroni adjustment applied as part of the test procedure.

The occurrence of each endpoint according to the etiologies was compared using Kaplan-Meier curves and the log-rank test. The incidence rate for each outcome was estimated per 100 patient-years. In the Cox proportional hazards models, baseline HRs were stratified according to the trial and with the treatment assignment as a covariate. Adjusted HRs were further adjusted for established cardiovascular risk factors: age, sex, race, heart rate, NYHA functional class III or IV, systolic blood pressure, body mass index, LVEF, estimated glomerular filtration rate, NT-proBNP (log transformed), history of atrial fibrillation, history of myocardial infarction, history of diabetes, implantable cardioverter-defibrillator use, and treatment setting (inpatient or outpatient). For analyses of stroke outcomes, additional adjustment for history of stroke and baseline use of anticoagulant therapy was performed as a sensitivity analysis. To assess whether the treatment effect for the primary outcome differed between patients with Chagasic and non-Chagasic patients, Cox proportional hazards models including a treatment-by-Chagas interaction term were used. Analyses were conducted in the pooled data set (stratified by trial) and separately within each trial, and interaction was assessed using likelihood ratio tests. We selected ischemic etiology as the reference group to compare with nonischemic etiologies. Additionally, adjusted HRs for clinical outcomes comparing Chagasic HF with other nonischemic etiologies were estimated using Cox proportional hazards models stratified by trial and adjusted for the aforementioned covariates. We did not adjust for geographic region in the primary analysis, because of concerns regarding potential collinearity with Chagasic etiology, which could complicate interpretation of conditional effects. As a sensitivity analysis, we also performed the analysis restricted to outpatients and to participants from Latin America, and additionally repeated the multivariable models with further adjustment for geographic region. In addition, the type of stroke was identified using the GALACTIC-HF data set, as it reported stroke types classified as ischemic, hemorrhagic, ischemic with hemorrhagic transformation, or undetermined.

For all analyses except those examining mode of death, we followed the original trial definitions of cardiovascular death. Accordingly, in ATMOSPHERE, deaths of unknown cause were included within cardiovascular death, consistent with the original publication. In contrast, in PARADIGM-HF and GALACTIC-HF, deaths of unknown cause were classified as a separate category and were not included within cardiovascular death. For analyses examining mode of death, unknown deaths were analyzed as a distinct category across all trials, including ATMOSPHERE, to ensure consistent classification. All statistical analyses were conducted using Stata version 18 (StataCorp), and a *P* value of <0.05 was considered to indicate statistical significance.

RESULTS

Overall, 23,647 patients were classified into etiologic categories: 13,381 ischemic (56.6%), 4,344 idiopathic (18.4%), 2,559 hypertensive (10.8%), 1,923 others (8.1%), 423 alcoholic (1.8%), 412 valvular (1.7%), 297 viral (1.3%), and 308 Chagasic (1.3%).

BASELINE CHARACTERISTICS OF CHAGASIC VS NON-CHAGASIC HF ETIOLOGIES. Baseline characteristics varied widely according to HF etiology (Table 1). Patients with Chagasic HF were most likely to be women (36.7%), and those with an alcoholic etiology were least likely to be women (3.1%). The next highest proportion of women was among patients with a valvular etiology (31.3%), and the second lowest proportion was among participants with ischemic heart disease (18.0%). Patients with Chagas disease had the lowest heart rate, blood pressure, and body mass index among all etiologies but had the highest median NT-proBNP level (with the next highest in patients with valvular disease). Patients with Chagas disease had few comorbidities compared with other groups, for example, diabetes (12.7% vs 40.3% in ischemic etiology) and atrial fibrillation

TABLE 1 Baseline Characteristics According to the Etiology in Patients (Overall) With Heart Failure With Reduced Ejection Fraction

	Chagasic (n = 308)	Ischemic (n = 13,381)	Valvular (n = 412)	Alcoholic (n = 423)	Hypertensive (n = 2,559)	Idiopathic (n = 4,344)	Viral (n = 297)	Others (n = 1,923)	*P* Value
Age, y	60.9 ± 10.9	66.0 ± 10.1	65.7 ± 11.6	56.5 ± 11.2	64.9 ± 11.4	60.2 ± 12.7	55.9 ± 13.4	58.9 ± 13.2	<0.001
Female	113 (36.7)	2,404 (18.0)	129 (31.3)	13 (3.1)	793 (31.0)	1,033 (23.8)	78 (26.3)	543 (28.2)	<0.001
Region									<0.001
North America	0 (0)	1,240 (9.3)	51 (12.4)	39 (9.2)	209 (8.2)	338 (7.8)	45 (15.2)	243 (12.6)	
Latin and Central America	307 (99.7)	1,717 (12.8)	73 (17.7)	62 (14.7)	885 (34.6)	812 (18.7)	25 (8.4)	245 (12.7)	
Western Europe	0 (0)	2,889 (21.6)	145 (35.2)	110 (26.0)	472 (18.4)	988 (22.7)	71 (23.9)	443 (23.0)	
Central and Eastern Europe (including Russia)	0 (0)	5,174 (38.7)	102 (24.8)	109 (25.8)	657 (25.7)	1,047 (24.1)	100 (33.7)	332 (17.3)	
Asia-Pacific and other	1 (0.3)	2,361 (17.6)	41 (10.0)	103 (24.4)	336 (13.1)	1,159 (26.7)	56 (18.9)	660 (34.3)	
Race									<0.001
White	168 (54.6)	10,140 (75.8)	324 (78.6)	266 (62.9)	1,642 (64.2)	2,664 (61.3)	224 (75.4)	1,105 (57.5)	
Black	53 (17.2)	294 (2.2)	21 (5.1)	46 (10.9)	283 (11.1)	258 (5.9)	16 (5.4)	128 (6.7)	
Asian	0 (0)	2,041 (15.3)	29 (7.0)	76 (18.0)	221 (8.6)	1,000 (23.0)	39 (13.1)	577 (30.0)	
Other	87 (28.3)	906 (6.8)	38 (9.2)	35 (8.3)	413 (16.1)	422 (9.7)	18 (6.1)	113 (5.9)	
Trial									<0.001
ATMOSPHERE	83 (27.0)	3,930 (29.4)	99 (24.0)	129 (30.5)	806 (31.5)	1,336 (30.8)	42 (14.1)	591 (30.7)	
PARADIGM-HF	113 (36.7)	5,036 (37.6)	110 (26.7)	158 (37.4)	968 (37.8)	1,595 (36.7)	164 (55.2)	255 (13.3)	
GALACTIC-HF	112 (36.4)	4,415 (33.0)	203 (49.3)	136 (32.2)	785 (30.7)	1,413 (32.5)	91 (30.6)	1,077 (56.0)	
Clinical characteristics									
NYHA functional class III/IV	56 (18.2)	5,239 (39.2)	157 (38.1)	119 (28.2)	857 (33.5)	1,289 (29.7)	80 (26.9)	723 (37.6)	<0.001
Inpatient	33 (10.7)	1,145 (8.6)	61 (14.8)	31 (7.3)	183 (7.2)	349 (8.0)	15 (5.1)	267 (13.9)	<0.001
KCCQ total symptom score	77.1 (60.4-90.6)	72.9 (54.7-87.5)	69.8 (52.9-89.6)	80.2 (64.6-92.7)	75.3 (56.3-89.6)	79.6 (60.2-92.7)	81.5 (64.1-93.8)	76.3 (58.3-90.6)	<0.001
Physiological and laboratory measurements									
Body mass index, kg/m^2	25.7 ± 4.6	28.0 ± 5.4	26.8 ± 5.5	27.4 ± 5.6	29.3 ± 6.1	27.7 ± 6.1	28.9 ± 5.6	27.7 ± 6.4	<0.001
Heart rate, beats/min	66.0 ± 10.2	71.4 ± 11.8	73.2 ± 13.1	75.1 ± 13.8	74.1 ± 12.7	72.9 ± 12.3	73.2 ± 12.2	73.6 ± 13.2	<0.001
Systolic blood pressure, mm Hg	109.1 ± 13.5	121.1 ± 16.2	115.3 ± 15.3	117.9 ± 15.7	126.5 ± 16.4	117.9 ± 16.4	117.7 ± 16.2	116.3 ± 16.7	<0.001
LVEF, %	27.2 ± 6.5	28.8 ± 6.0	27.0 ± 7.0	25.8 ± 6.1	28.9 ± 6.1	26.8 ± 6.2	26.9 ± 6.7	26.7 ± 6.2	<0.001
NT-proBNP, pg/mL	2,286 (1,042-4,463)	1,553 (812-3,077)	2,137 (1,148-3,848)	1,644 (881-3,645)	1,693 (888-3,471)	1,566 (812-3,287)	1,639 (754-3,566)	1,609 (835-3,182)	<0.001
Hemoglobin, g/dL	13.7 ± 1.6	13.7 ± 1.7	13.4 ± 1.6	14.3 ± 1.6	14.0 ± 1.7	13.9 ± 1.7	14.1 ± 1.6	13.9 ± 1.7	<0.001
Serum creatinine, μmol/L	105.5 ± 44.2	105.5 ± 33.7	105.9 ± 34.9	94.9 ± 25.3	98.5 ± 31.0	98.0 ± 31.0	100.4 ± 32.5	99.1 ± 34.0	<0.001
eGFR, mL/min/1.73 m^2	65.2 ± 22.4	64.6 ± 20.7	64.8 ± 42.1	78.2 ± 22.5	68.3 ± 21.3	71.0 ± 23.3	69.2 ± 21.5	70.9 ± 26.8	<0.001
Medical history									
Diabetes mellitus	39 (12.7)	5,386 (40.3)	97 (23.5)	91 (21.5)	823 (32.2)	1,132 (26.1)	73 (24.6)	587 (30.5)	<0.001
Hypertension	129 (41.9)	9,769 (73.0)	210 (51.0)	215 (50.8)	2,491 (97.3)	2,148 (49.5)	137 (46.1)	957 (49.8)	<0.001
Myocardial infarction	4 (1.3)	9,505 (71.0)	23 (5.6)	15 (3.6)	132 (5.2)	145 (3.3)	6 (2.0)	86 (4.5)	<0.001
AF or atrial flutter at screening	110 (35.7)	4,793 (35.8)	259 (62.9)	176 (41.6)	1,132 (44.2)	1,590 (36.6)	104 (35.0)	792 (41.2)	<0.001
Stroke	37 (12.1)	1,309 (9.8)	45 (10.9)	21 (5.0)	183 (7.2)	253 (5.8)	13 (4.4)	110 (5.7)	<0.001
Treatments									
ACEI, ARB, or ARNI[a]	289 (94.2)	12,768 (95.4)	372 (90.3)	407 (96.2)	2,487 (97.2)	4,171 (96.0)	292 (98.3)	1,787 (92.9)	<0.001
BB	269 (87.3)	12,460 (93.1)	380 (92.2)	398 (94.1)	2,389 (93.4)	4,086 (94.1)	274 (92.3)	1,751 (91.1)	<0.001
SGLT2 inhibitor	0 (0)	124 (0.9)	4 (1.0)	2 (0.5)	7 (0.3)	52 (1.2)	1 (0.3)	28 (1.5)	<0.001
MRAs	234 (76.0)	7,311 (54.6)	270 (65.5)	286 (67.6)	1,422 (55.6)	2,697 (62.1)	185 (62.3)	1,265 (65.8)	<0.001
Triple therapy (ACEI/ARB/ARNI + BB + MRA)	194 (63.0)	6,548 (48.9)	222 (53.9)	261 (61.7)	1,310 (51.2)	2,439 (56.2)	176 (59.3)	1,096 (57.0)	<0.001
Anticoagulation	128 (41.6)	6,752 (50.5)	319 (77.4)	194 (45.9)	1,208 (47.2)	1,949 (44.9)	130 (43.8)	1,022 (53.1)	<0.001
Antiplatelet agents	85 (27.6)	9,238 (69.0)	106 (25.7)	131 (31.0)	1,040 (40.6)	1,494 (34.4)	94 (31.7)	625 (32.5)	<0.001
Diuretic agents	258 (83.8)	10,958 (81.9)	359 (87.1)	366 (86.5)	2,198 (85.9)	3,693 (85.0)	241 (81.1)	1,641 (85.3)	<0.001
Pacemaker	85 (27.6)	1,219 (9.1)	80 (19.4)	22 (5.2)	160 (6.3)	410 (9.4)	25 (8.4)	129 (6.7)	<0.001
Cardiac resynchronization therapy	13 (4.2)	1,180 (8.8)	76 (18.5)	29 (6.9)	93 (3.6)	500 (11.5)	39 (13.1)	195 (10.1)	<0.001
Implantable cardioverter-defibrillator	43 (14.0)	3,122 (23.3)	106 (25.7)	60 (14.2)	221 (8.6)	873 (20.1)	72 (24.2)	408 (21.2)	<0.001

Values are mean ± SD, n (%), or median (Q1-Q3). Baseline characteristics were referred at the randomization of each group. [a]In PARADIGM-HF and ATMOSPHERE, all patients were regarded as ACEI, ARB, or ARNI use on the basis of allocated treatment.

ACEI = angiotensin-converting enzyme inhibitor; AF = atrial fibrillation; ARB = angiotensin receptor blocker; ARNI = angiotensin receptor neprilysin inhibitor; ATMOSPHERE = Aliskiren Trial to Minimize Outcomes in Patients With Heart Failure; BB = beta-blocker; eGFR = estimated glomerular filtration rate; GALACTIC-HF = Global Approach to Lowering Adverse Cardiac Outcomes Through Improving Contractility in Heart Failure; KCCQ = Kansas City Cardiomyopathy Questionnaire; LVEF = left ventricular ejection fraction; MRA = mineralocorticoid receptor antagonist; NT-proBNP = N-terminal pro-B-type natriuretic peptide; PARADIGM-HF = Prospective Comparison of ARNI With ACEI to Determine Impact on Global Mortality and Morbidity in Heart Failure; SGLT2i = sodium-glucose cotransporter 2 inhibitor.

(35.7% vs 62.9% in valvular etiology), but had the most frequent history of stroke (12.1% vs the lowest rate of 4.4% among participants with a viral etiology). Patients with Chagasic HF were least likely to be in NYHA functional class III or IV (18.2%), and participants with a valvular etiology were most likely to have this more severe limitation (38.1%). In terms of treatment, the use of triple therapy (angiotensin-converting enzyme inhibitor, angiotensin receptor blocker, or angiotensin receptor neprilysin inhibitor; beta-blocker; and mineralocorticoid receptor antagonist) and pacemaker implantation were most frequent in Chagasic HF, whereas implantable cardioverter-defibrillator use was less common (most frequent in patients with an ischemic etiology). As enrollment in these studies largely predated the widespread adoption of sodium-glucose cotransporter 2 inhibitors, their use was minimal. The use of anticoagulation was lowest among patients with Chagasic HF (41.6%) and highest among those with a valvular etiology (77.4%).

CLINICAL OUTCOMES. Clinical outcomes according to etiology are presented in **Tables 2 and 3**, Supplemental Table 2, **Figures 1 and 2**, and the **Central Illustration**. Despite their younger age and lack of comorbidities, patients with Chagasic HF had higher rates of all clinical outcomes compared with those with an ischemic etiology (**Table 2**). Compared with other nonischemic etiologies, Chagasic HF showed elevated HRs (>1.0) across all outcomes, the majority of which were statistically significant (**Table 3**). Patients with valvular and viral etiologies were also notable for their high and low rates of events, respectively.

Primary composite outcome. Patients with Chagas disease had by far the highest incidence of the primary outcome (HF hospitalization or cardiovascular death): rates per 100 patient-years were 22.9 (95% CI: 19.1-27.4) for Chagasic, 16.0 (95% CI: 15.5-16.5) for ischemic, 16.5 (95% CI: 14.0-19.5) for valvular, 13.7 (95% CI: 11.5-16.3) for alcoholic, 13.1 (95% CI: 12.1-14.1) for hypertensive, 13.7 (95% CI: 12.9-14.4) for idiopathic, 12.3 (95% CI: 9.8-15.3) for viral, and 14.5 (95% CI: 13.4-15.8) for "other" (**Table 2**, **Figure 1**). In the adjusted model, the HRs for the primary outcome were 1.65 (95% CI: 1.36-2.02) for Chagasic, 0.91 (95% CI: 0.76-1.09) for valvular, 0.95 (95% CI: 0.79-1.15) for alcoholic, 0.94 (95% CI: 0.85-1.03) for hypertensive, 1.01 (95% CI: 0.93-1.09) for idiopathic, 0.98 (95% CI: 0.78-1.23) for viral, and 0.92 (95% CI: 0.84-1.02) for others, compared with those with ischemic HF (**Table 2**). Similar trends for clinical outcomes were observed when the analysis was confined to outpatients only and to participants from Latin America (Supplemental Tables 3-6). For the primary outcome, there was no evidence that the treatment effect differed between patients with Chagasic and non-Chagasic cardiomyopathy in the pooled analysis (P for interaction = 0.83). Trial-specific analyses likewise demonstrated no significant treatment-by-Chagas interaction in ATMOSPHERE (P for interaction = 0.12), GALACTIC-HF (P for interaction = 0.85), or PARADIGM-HF (P for interaction = 0.57).

Sudden and pump failure death. The crude incidence of each of pump failure (or worsening HF) death and sudden death was higher in patients with Chagasic HF than in any other etiologic group. The second highest incidence of sudden death was among patients with an ischemic etiology, and the second highest incidence of pump failure death among those with a valvular etiology; the lowest risk for both types of death was in participants with a viral etiology.

The risk (HR) for pump failure death was significantly higher among patients with Chagas disease than for any other etiology. Although the HR for sudden death was also higher among patients with a Chagasic etiology compared with others, the evidence for this association was weaker, although the overall number of sudden deaths among patients with Chagas disease was small (**Tables 2 and 3**).

As a proportion of all deaths, sudden death was smaller in patients with Chagas disease (21.2%) (and valvular disease [20.0%]) than among the other etiologies (eg, 29%-30% for ischemic and hypertensive HFrEF) (**Figure 3**, Supplemental Table 7).

All-cause death. The incidence of death of any cause was much higher among patents with Chagasic HF than in any other etiologic group. The risk (HR) for all-cause death was significantly higher among patients with Chagas disease than for any other etiology, approximately double that of any other group and triple that of patients with a viral etiology (**Tables 2 and 3**).

HF hospitalization. The risk for HF hospitalization was higher among patients with Chagas disease compared with other causes, except for viral etiology, although the competing risk for death was much smaller in that group.

Stroke. Chagasic HF was associated with a significantly higher risk for stroke (adjusted HR range: 1.89-5.71) compared with other nonischemic etiologies, except for valvular HF (HR: 1.54; 95% CI: 0.73-3.23). After additional adjustment for history of stroke and baseline use of anticoagulant therapy, the associations were attenuated; however, the overall pattern

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TABLE 2 Clinical Outcomes According to Etiology

	Chagasic (n = 308 [1.3%])	Ischemic (n = 13,381 [56.6%])	Valvular (n = 412 [1.7%])	Alcoholic (n = 423 [1.8%])
HF hospitalization or CV death				
Number of events (%)	119 (38.6)	4,420 (33.0)	136 (33.0)	127 (30.0)
Event rate (95% CI)	22.9 (19.1-27.4)	16.0 (15.5-16.5)	16.5 (14.0-19.5)	13.7 (11.5-16.3)
HR (95% CI)[a]	1.36 (1.13-1.63)	Reference	0.94 (0.79-1.11)	0.86 (0.72-1.03)
Adjusted HR (95% CI)[b]	1.65 (1.36-2.02)	Reference	0.91 (0.76-1.09)	0.95 (0.79-1.15)
CV death				
Number of events (%)	84 (27.3)	2,685 (20.1)	83 (20.2)	72 (17.0)
Event rate (95% CI)	14.4 (11.7-17.9)	8.7 (8.4-9.0)	9.0 (7.3-11.1)	7.1 (5.7-9.0)
HR (95% CI)[a]	1.65 (1.33-2.05)	Reference	0.97 (0.78-1.20)	0.82 (0.65-1.03)
Adjusted HR (95% CI)[b]	1.86 (1.47-2.35)	Reference	0.89 (0.71-1.12)	0.88 (0.68-1.12)
All-cause death				
Number of events (%)	104 (33.8)	3,364 (25.1)	100 (24.3)	92 (21.8)
Event rate (95% CI)	17.9 (14.7-21.7)	10.9 (10.5-11.3)	10.8 (8.9-13.2)	9.1 (7.4-11.2)
HR (95% CI)[a]	1.65 (1.36-2.01)	Reference	0.92 (0.75-1.12)	0.84 (0.68-1.03)
Adjusted HR (95% CI)[b]	1.82 (1.47-2.25)	Reference	0.84 (0.68-1.04)	0.91 (0.73-1.14)
HF hospitalization				
Number of events (%)	74 (24.0)	2,797 (20.9)	99 (24.0)	84 (19.9)
Event rate (95% CI)	14.2 (11.3-17.9)	10.1 (9.8-10.5)	12.0 (9.9-14.6)	9.1 (7.3-11.2)
HR (95% CI)[a]	1.31 (1.04-1.65)	Reference	1.03 (0.84-1.26)	0.91 (0.73-1.12)
Adjusted HR (95% CI)[b]	1.75 (1.36-2.25)	Reference	1.04 (0.84-1.28)	1.08 (0.86-1.35)
Sudden cardiac death				
Number of events (%)	22 (7.1)	987 (7.4)	20 (4.9)	29 (6.9)
Event rate (95% CI)	3.8 (2.5-5.7)	3.2 (3.0-3.4)	2.2 (1.4-3.4)	2.9 (2.0-4.1)
HR (95% CI)[a]	1.16 (0.76-1.77)	Reference	0.70 (0.45-1.08)	0.90 (0.62-1.30)
Adjusted HR (95% CI)[b]	1.16 (0.73-1.84)	Reference	0.72 (0.46-1.14)	0.81 (0.55-1.18)
Pump failure death				
Number of events (%)	40 (13.0)	857 (6.4)	44 (10.7)	24 (5.7)
Event rate (95% CI)	6.9 (5.0-9.4)	2.8 (2.6-3.0)	4.8 (3.5-6.4)	2.4 (1.6-3.5)
HR (95% CI)[a]	2.45 (1.78-3.36)	Reference	1.46 (1.08-1.98)	0.86 (0.57-1.29)
Adjusted HR (95% CI)[b]	2.82 (1.98-4.03)	Reference	1.18 (0.86-1.64)	1.13 (0.74-1.73)
Non-CV death				
Number of events (%)	18 (5.8)	541 (4.0)	12 (2.9)	16 (3.8)
Event rate (95% CI)	3.1 (1.9-4.9)	1.8 (1.6-1.9)	1.3 (0.7-2.3)	1.6 (1.0-2.6)
HR (95% CI)[a]	1.84 (1.15-2.95)	Reference	0.67 (0.38-1.19)	0.90 (0.55-1.48)
Adjusted HR (95% CI)[b]	2.09 (1.25-3.49)	Reference	0.61 (0.34-1.09)	1.16 (0.69-1.95)
Stroke (fatal or nonfatal)				
Number of events (%)	15 (4.9)	409 (3.1)	16 (3.9)	8 (1.9)
Event rate (95% CI)	2.6 (1.6-4.4)	1.3 (1.2-1.5)	1.8 (1.1-2.9)	0.8 (0.4-1.6)
HR (95% CI)[a]	1.92 (1.15-3.21)	Reference	1.31 (0.79-2.15)	0.59 (0.29-1.20)
Adjusted HR (95% CI)[b]	2.16 (1.20-3.88)	Reference	1.40 (0.83-2.36)	0.62 (0.29-1.34)

Event rate is the number of events per 100 person-years. [a]Stratified according to the trial and adjusted for treatment assignment. [b]Further adjusted for age, sex, race, heart rate, NYHA functional class III or IV, systolic blood pressure, body mass index, left ventricular ejection fraction, estimated glomerular filtration rate, N-terminal pro-B-type natriuretic peptide (log transformed), history of atrial fibrillation, history of myocardial infarction, history of diabetes, implantable cardioverter-defibrillator use, and treatment setting (inpatient or outpatient).

CV = cardiovascular; HF = heart failure.

Continued on the next page

remained similar, with higher stroke risk observed in Chagasic HF compared with most etiologies (adjusted HR range: 2.08-5.35), except for valvular and “other” etiologies (Supplemental Table 8). The most common type of stroke in GALACTIC-HF was ischemic stroke across all etiologies (Supplemental Table 9). There was no significant difference in the incidence of new-onset atrial fibrillation in patients with Chagas HF compared with other etiologies (Supplemental Table 10).

TABLE 2 Continued

Hypertensive (n = 2,559 [10.8%])	Idiopathic (n = 4,344 [18.4%])	Viral (n = 297 [1.3%])	Others (n = 1,923 [8.1%])
713 (27.9)	1,283 (29.5)	79 (26.6)	579 (30.1)
13.1 (12.1-14.1)	13.7 (12.9-14.4)	12.3 (9.8-15.3)	14.5 (13.4-15.8)
0.83 (0.76-0.90)	0.86 (0.81-0.92)	0.79 (0.63-0.99)	0.79 (0.72-0.86)
0.94 (0.85-1.03)	1.01 (0.93-1.09)	0.98 (0.78-1.23)	0.92 (0.84-1.02)
438 (17.1)	772 (17.8)	33 (11.1)	311 (16.2)
7.3 (6.7-8.0)	7.4 (6.9-7.9)	4.6 (3.3-6.5)	7.0 (6.3-7.8)
0.85 (0.77-0.94)	0.85 (0.78-0.92)	0.54 (0.39-0.77)	0.73 (0.65-0.82)
0.95 (0.84-1.07)	0.95 (0.86-1.05)	0.67 (0.47-0.95)	0.84 (0.74-0.96)
534 (20.9)	932 (21.5)	41 (13.8)	406 (21.1)
8.9 (8.2-9.7)	8.9 (8.4-9.5)	5.7 (4.2-7.8)	9.1 (8.3-10.1)
0.83 (0.75-0.91)	0.82 (0.76-0.88)	0.53 (0.39-0.73)	0.75 (0.68-0.84)
0.90 (0.81-1.01)	0.92 (0.84-1.01)	0.66 (0.48-0.90)	0.87 (0.78-0.98)
424 (16.6)	885 (20.4)	65 (21.9)	412 (21.4)
7.8 (7.1-8.5)	9.4 (8.8-10.1)	10.1 (7.9-12.9)	10.4 (9.4-11.4)
0.78 (0.71-0.87)	0.94 (0.87-1.02)	1.03 (0.81-1.32)	0.85 (0.76-0.94)
0.94 (0.83-1.06)	1.15 (1.05-1.27)	1.29 (1.00-1.66)	1.02 (0.91-1.15)
158 (6.2)	260 (6.0)	9 (3.0)	95 (4.9)
2.6 (2.3-3.1)	2.5 (2.2-2.8)	1.3 (0.7-2.4)	2.1 (1.7-2.6)
0.82 (0.69-0.97)	0.78 (0.68-0.89)	0.40 (0.21-0.77)	0.69 (0.56-0.86)
0.86 (0.71-1.04)	0.77 (0.65-0.91)	0.44 (0.23-0.85)	0.67 (0.53-0.85)
121 (4.7)	309 (7.1)	16 (5.4)	128 (6.7)
2.0 (1.7-2.4)	3.0 (2.6-3.3)	2.2 (1.4-3.6)	2.9 (2.4-3.4)
0.74 (0.62-0.90)	1.08 (0.94-1.23)	0.83 (0.50-1.35)	0.84 (0.69-1.01)
1.00 (0.81-1.25)	1.28 (1.09-1.52)	1.02 (0.61-1.70)	1.08 (0.87-1.34)
69 (2.7)	121 (2.8)	5 (1.7)	71 (3.7)
1.2 (0.9-1.5)	1.2 (1.0-1.4)	0.7 (0.3-1.7)	1.6 (1.3-2.0)
0.67 (0.52-0.86)	0.67 (0.55-0.81)	0.40 (0.17-0.97)	0.80 (0.63-1.04)
0.68 (0.52-0.91)	0.78 (0.61-0.98)	0.52 (0.21-1.28)	0.98 (0.73-1.31)
76 (3.0)	103 (2.4)	3 (1.0)	60 (3.1)
1.3 (1.0-1.6)	1.0 (0.8-1.2)	0.4 (0.1-1.3)	1.4 (1.1-1.8)
0.95 (0.74-1.21)	0.74 (0.59-0.91)	0.32 (0.10-1.00)	1.00 (0.76-1.32)
0.91 (0.68-1.22)	0.80 (0.61-1.05)	0.38 (0.12-1.19)	1.14 (0.83-1.56)

DISCUSSION

In this large, patient-level analysis combining 3 major global randomized trials of HFrEF, we provide the most comprehensive comparison to date of Chagasic HF with HFrEF due to other causes, including ischemic heart disease. Our analysis also provides the most contemporary comparison of outcomes among patients with HFrEF due to a range of nonischemic etiologies. There are several key findings.

First, Chagasic HF exhibited a substantially worse prognosis than all other nonischemic etiologies and even HF due to an ischemic cause. This excess risk persisted after multivariable adjustment. Second, risk, including risk for death, varied among the other nonischemic etiologies. After Chagasic HF, event rates were highest in patients with a valvular etiology, which approached those observed in patients with ischemic heart disease (despite the exclusion of severe uncorrected primary valvular disease in all

TABLE 3 Clinical Outcomes Comparing Chagasic and Other Nonischemic Etiologies

	Adjusted HR[a]					
	vs Valvular	vs Alcoholic	vs Hypertensive	vs Idiopathic	vs Viral	vs Others
HF hospitalization or CV death	1.82 (1.41-2.35)	1.74 (1.34-2.25)	1.77 (1.44-2.16)	1.64 (1.35-2.00)	1.69 (1.26-2.26)	1.79 (1.45-2.20)
Cardiovascular death	2.08 (1.52-2.85)	2.12 (1.53-2.94)	1.96 (1.53-2.50)	1.95 (1.54-2.47)	2.79 (1.85-4.19)	2.20 (1.71-2.84)
All-cause death	2.16 (1.63-2.87)	1.99 (1.49-2.66)	2.01 (1.62-2.51)	1.97 (1.60-2.44)	2.76 (1.91-3.98)	2.08 (1.66-2.61)
HF hospitalization	1.69 (1.23-2.31)	1.63 (1.17-2.25)	1.86 (1.43-2.42)	1.52 (1.18-1.95)	1.36 (0.96-1.92)	1.71 (1.31-2.22)
Sudden cardiac death	1.60 (0.86-3.00)	1.44 (0.81-2.56)	1.35 (0.84-2.17)	1.50 (0.95-2.39)	2.64 (1.20-5.83)	1.73 (1.06-2.83)
Pump failure death	2.39 (1.53-3.73)	2.49 (1.48-4.19)	2.81 (1.93-4.10)	2.20 (1.55-3.11)	2.76 (1.53-4.99)	2.62 (1.80-3.80)
Non-CV death	3.43 (1.62-7.24)	1.80 (0.90-3.59)	3.05 (1.77-5.25)	2.69 (1.60-4.52)	3.98 (1.46-10.87)	2.13 (1.23-3.68)
Stroke (fatal or nonfatal)	1.54 (0.74-3.23)	3.46 (1.37-8.75)	2.36 (1.29-4.32)	2.68 (1.48-4.85)	5.71 (1.62-20.15)	1.89 (1.02-3.49)

HRs were derived from the same multivariable model as in Table 2. For each column, the reference group was set to the etiology indicated in the column header. [a]Stratified according to the trial and adjusted for treatment assignment, age, sex, race, heart rate, NYHA functional class III or IV, systolic blood pressure, body mass index, left ventricular ejection fraction, estimated glomerular filtration rate, N-terminal pro-B-type natriuretic peptide (log transformed), history of atrial fibrillation, history of myocardial infarction, history of diabetes, implantable cardioverter-defibrillator use, and treatment setting (inpatient or outpatient).

Abbreviations as in Table 2.

trials). Conversely, patients with a viral etiology had the lowest event rates. Third, the risk for stroke was significantly higher in Chagasic HF than ischemic and other nonischemic etiologies, although the incidence of stroke was also notable in patients with a valvular etiology.

Approximately one-half the patients with HFrEF in this study had an investigator-reported nonischemic etiology, of which 3% were attributed to Chagas disease. To the best of our knowledge, this is the first study to compare outcomes between Chagasic HF and other nonischemic etiologies, such as valvular, alcoholic, and viral cardiomyopathy, as well as an ischemic cause. The latter is consistently associated with a worse prognosis and with a higher risk for sudden death than nonischemic HFrEF. Yet patients with HFrEF due to Chagas disease had a higher incidence of all outcomes examined, including sudden death and all-cause mortality, compared with patients with an ischemic etiology. The HR for death of any cause was 1.82 (95% CI: 1.47-2.25) among patients with Chagas disease compared with those with an ischemic etiology, and the HR for this outcome was consistently about 2.0 across the range of other nonischemic etiologies investigated. The explanation for the very high mortality rate in patients with HF due to Chagas disease is uncertain, but it has been attributed to persistent immune-mediated myocardial inflammation due to chronic parasitic infection (although the importance of these mechanisms in all patients is debated), reduced right ventricular function, microvascular and autonomic dysfunction, a hypercoagulable state, ventricular aneurysm formation, and elevated risk for stroke, as well as conduction disturbances and ventricular arrhythmias leading to a high risk for sudden death.[17-19] Although we confirmed that the incidence of sudden death was high in patients with Chagasic HF, we found that this was even more so for death due to pump failure and that the proportion of sudden deaths was not as high in patients with Chagas disease as in other etiologies (eg, ischemic heart disease).

There have been few comparisons of outcomes among patients with cardiomyopathy due to different etiologies since the seminal study of Felker et al[20] published in 2000. In that report, the adjusted HR for all-cause mortality in patients with an ischemic etiology ($n = 91$) was 1.52 (95% CI: 1.07-2.17) compared with an idiopathic etiology (reference group $n = 616$; $P = 0.02$); mortality in patients with cardiomyopathy due to infiltrative myocardial disease (ie, amyloidosis, hemochromatosis, or sarcoidosis; $n = 59$), infection with HIV ($n = 45$), and therapy with doxorubicin ($n = 15$) was also higher than in "idiopathic" cases. Mortality in patients with cardiomyopathy due to myocarditis ($n = 111$), substance abuse ($n = 37$; mainly alcohol abuse), connective tissue disease ($n = 39$), hypertension ($n = 49$), and "other" causes ($n = 117$) did not differ from "idiopathic" etiology. Conversely, participants with peripartum cardiomyopathy ($n = 51$) had a significantly better outcome than the reference group. Where this study and the present report overlap (idiopathic, ischemic, hypertensive, and alcoholic), the findings are largely congruent. However, Felker et al[20] did not have a valvular group that had a surprisingly poor prognosis in the present study or a viral group that had the best outcome of any etiology, 2 novel findings with clinical implications and worthy of further investigation.

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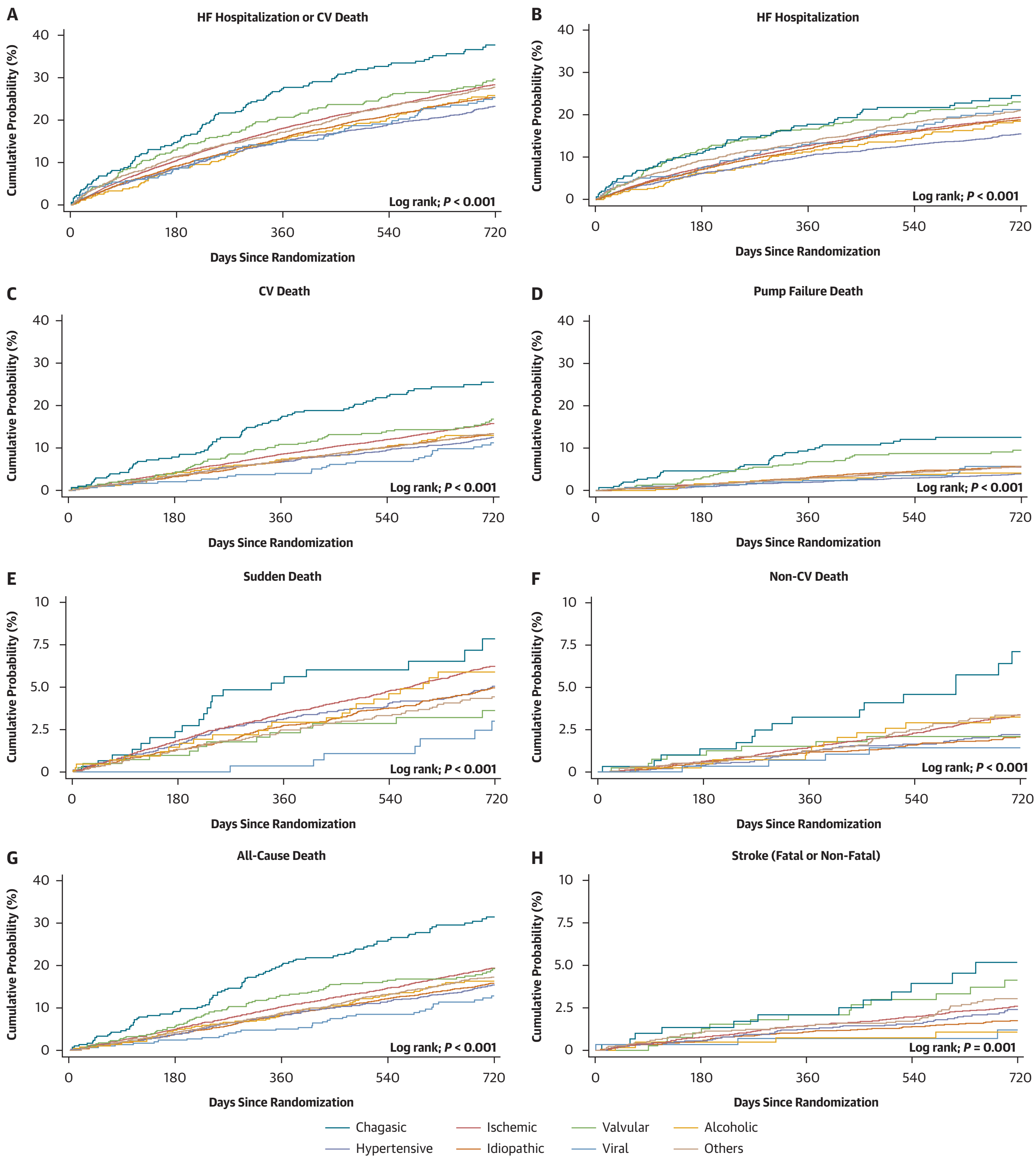
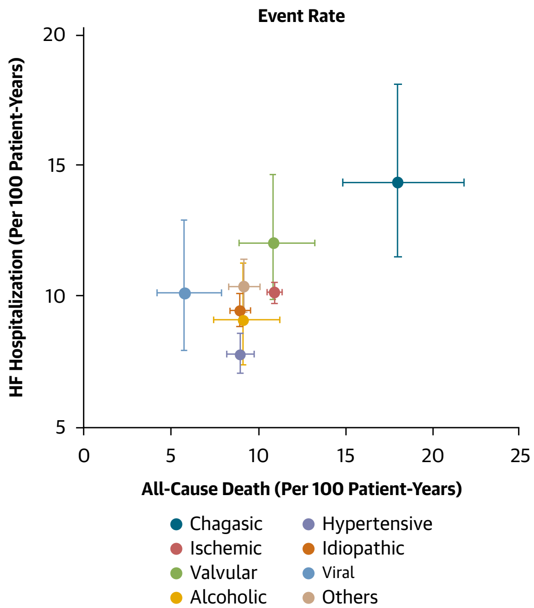
FIGURE 1 Kaplan-Meier Curves for Clinical Outcomes According to the Etiology in Heart Failure With Reduced Ejection Fraction (Overall)



(A) Heart failure (HF) hospitalization or cardiovascular (CV) death, (B) HF hospitalization, (C) CV death, (D) pump failure death, (E) sudden death, (F) non-CV death, (G) all-cause death, and (H) stroke (fatal or nonfatal).

FIGURE 2 Event Rate of All-Cause Death and HF Hospitalization According to Etiology in HF With Reduced Ejection Fraction



The events rates are calculated as the number of events per 100 patient-years of follow-up. The error bars indicate the 95% CI for each outcome. HF = heart failure.

Patients with Chagasic HF also had a considerably higher incidence of stroke compared with other HF etiologies. Compared with all other etiologies excluding valvular and the "other" etiology, Chagasic cardiomyopathy was associated with a 2.1- to 5.4-fold higher adjusted risk for stroke. This magnitude of excess risk is clinically substantial and underscores the importance of stroke as a major complication of Chagasic HF. Notably, the high risk for stroke was not explained by a more frequent history of atrial fibrillation or hypertension (patients with Chagas disease had the second lowest prevalence of atrial fibrillation and the lowest baseline blood pressure among the various etiologies). Furthermore, new-onset atrial fibrillation was not more frequent in patients with Chagasic HF compared with other etiologies. However, one possible explanation for the observed stroke risk may be regional differences in health care access and management. Access to and use of direct oral anticoagulant agents in Latin America have been reported to be lower than in other regions, as shown in real-world analyses from the GARFIELD-AF (Global Anticoagulant Registry in the Field) and GLORIA-AF (Global Registry on Long-Term Oral Antithrombotic Treatment in Patients With Atrial Fibrillation) registries, with correspondingly higher stroke rates.[21,22] A similar pattern has been described regarding blood pressure control in hypertension in Latin America, although the proportion of patients with histories of hypertension was low compared with other HFrEF etiologies, and the mean baseline systolic blood pressure in patients with Chagas disease was only 109 mm Hg.[23] Given that 307 of the 308 patients with Chagasic HFrEF in our analysis were enrolled from Latin America, and in the absence of data on direct oral anticoagulant use or blood pressure control for adjustment, regional differences in anticoagulation practices and other health care-related factors may have contributed to the observed excess stroke risk. Another possible explanation is that left ventricular apical aneurysm formation is pathognomonic of Chagas disease and a probable source of cardioembolism in these patients.[24,25] In addition, Chagas disease induces a systemic prothrombotic milieu characterized by chronic inflammation, endothelial dysfunction, and platelet activation, which may predispose to cerebrovascular events.[24] These observations suggest that exploration of the potential benefit of antithrombotic therapy, antiplatelet therapy, or both, in a randomized trial is worthwhile in Chagas cardiomyopathy.

Interestingly, we found that investigator-reported valvular etiology was also associated with a higher incidence of stroke than most other causes of HF. Although we do not know the specific valvular disease in these patients, the trials analyzed excluded participants with severe uncorrected primary valve disease, and most patients in this group likely had undergone prosthetic valve replacement. Although the prevalence of atrial fibrillation was very high in this group (77%), prosthetic valves are associated with an elevated risk for cardioembolic stroke, and patients with valvular disease have a higher risk for stroke even in the absence of atrial fibrillation.[26] In addition, potential overlap with rheumatic heart disease may also contribute to the observed excess risk for stroke in this group.

Although the present study has shown perhaps unexpected variation in clinical characteristics and outcomes among different etiologies in patients with HFrEF, the primary focus was on Chagasic HF as a neglected problem increasing in importance because of geographic spread. Our findings clearly demonstrate the uniquely poor outlook for these patients and reiterate the need to find effective therapies. The high event rates observed in Chagasic HF in the present analysis were consistent with those in 2 recent trials (PARACHUTE-HF [Efficacy and Safety of

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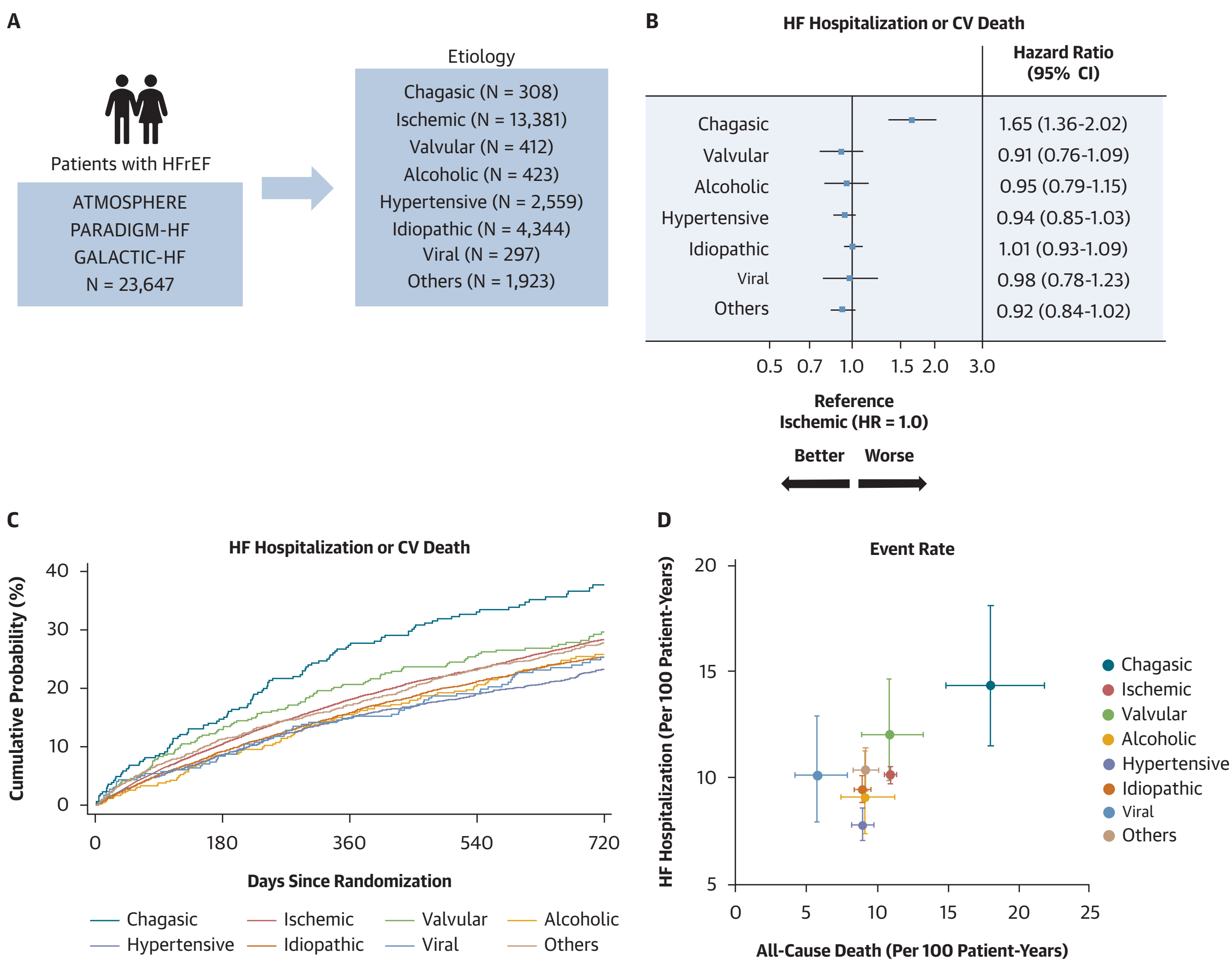
CENTRAL ILLUSTRATION Comparative Prognosis of Chagas and Other Cardiomyopathies



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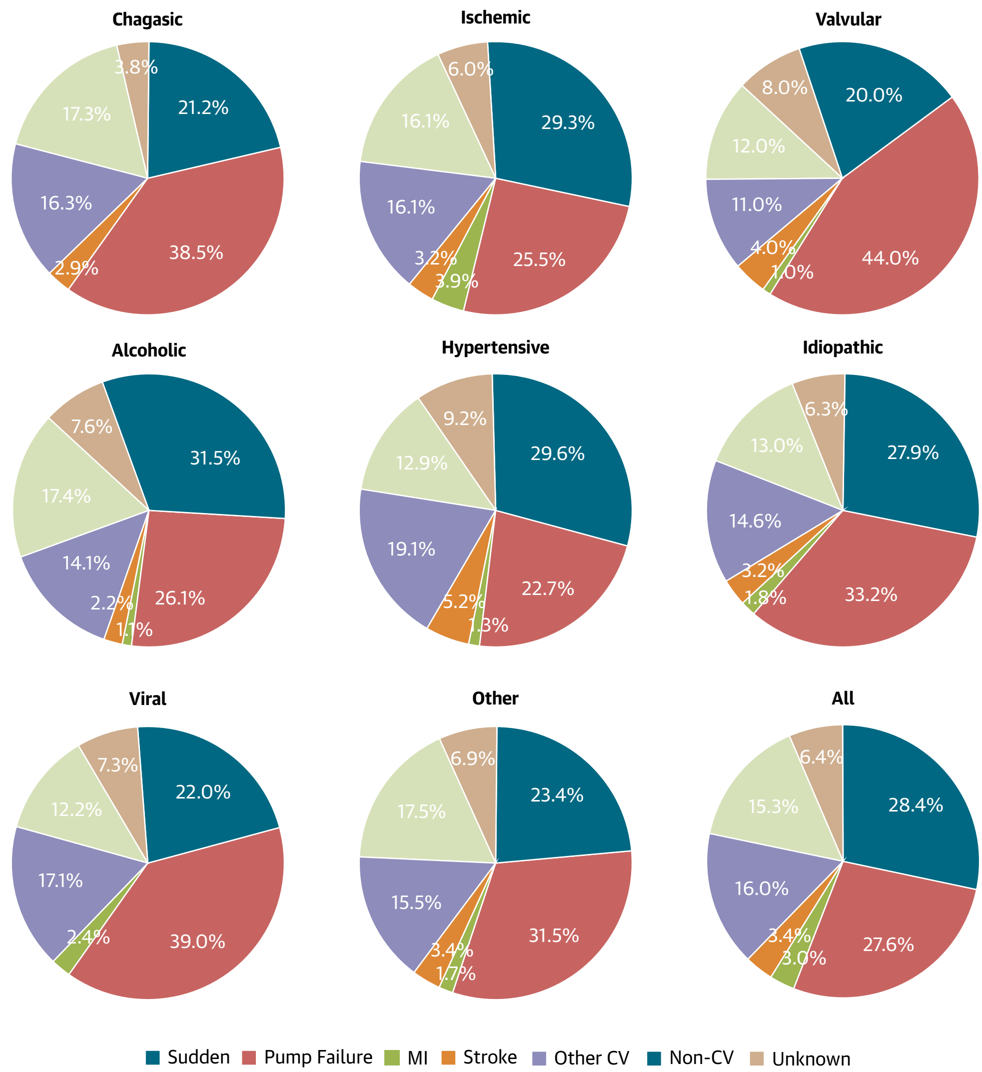
(A) Included trials and distribution of etiologies in heart failure with reduced ejection fraction (HFrEF). (B) Forest plot showing HRs (95% CIs) for the primary outcome (heart failure [HF] hospitalization or cardiovascular [CV] death) in nonischemic etiologies. Ischemic etiology served as the reference (HR: 1), and other nonischemic etiologies were compared. (C) Kaplan-Meier curves for the primary outcome (HF hospitalization or CV death). (D) Event rate of all-cause death and HF hospitalization according to etiology. ATMOSPHERE = Aliskiren Trial to Minimize Outcomes in Patients With Heart Failure; GALACTIC-HF = Global Approach to Lowering Adverse Cardiac Outcomes Through Improving Contractility in Heart Failure; PARADIGM-HF = Prospective Comparison of ARNI With ACEI to Determine Impact on Global Mortality and Morbidity in Heart Failure.

Sacubitril/Valsartan Compared With Enalapril on Morbidity, Mortality, and NT-proBNP Change in Patients With CCC] and ANSWER-HF [Angiotensin Receptor-Neprilysin Inhibition in Chagas Cardiomyopathy With Reduced Ejection Fraction]) enrolling patients with CCC and HFrEF, although the latter was relatively small and short term, with few events.[19,27]

Although patients with Chagas disease in the present cohort were broadly similar in terms of clinical characteristics to those in PARACHUTE-HF, event rates were somewhat higher in the present study compared with PARACHUTE-HF; for example, for the composite of first HF hospitalization or cardiovascular death, the rate was 22.9 (95% CI: 19.1-27.4) per

FIGURE 3 Mode of Death According to Etiology



Sudden death, pump failure death, death due to myocardial infarction (MI), death due to stroke, other cardiovascular (CV) death, non-CV death, and death of unknown cause are presented for each etiology.

100 patient-years in the present study compared with 18.8 per 100 patient-years in the enalapril group in PARACHUTE-HF. Similarly, the rate of all-cause mortality was higher in the present cohort than in PARACHUTE-HF: 17.9 (95% CI: 14.7-21.7) per 100 patient-years compared with 13.5 per 100 patient-years in the enalapril group of PARACHUTE-HF. This likely reflects the lower average LVEF (27.2% vs 29.8%) and higher median NT-proBNP concentration (2,286 vs 1,730 pg/mL) in the present cohort, which reflects the different enrollment criteria for the various trials vs PARACHUTE-HF.

Our study extends the findings of recent trials such as PARACHUTE-HF and ANSWER-HF, which focused exclusively on Chagasic cardiomyopathy, and our previous report,[12] by demonstrating that: 1) Chagasic HFrEF remains associated with excess risk despite contemporary therapy; 2) this risk exceeds that of other HF etiologies; and 3) stroke also represents a particularly important component of this excess risk. Importantly, these new trials (PARACHUTE-HF and ANSWER-HF) demonstrated the feasibility of testing therapeutic approaches for CCC, including those targeted at persistent infection, inflammation, conduction system disease, and the high risk for thromboembolism faced by these patients.[17-19]

STUDY LIMITATIONS. First, this study represents a post hoc analysis, which may be subject to residual confounding and hypothesis-generating bias. Second, the analyzed data were derived from patients enrolled in randomized controlled trials with specific inclusion and exclusion criteria, and the findings may not be generalizable to the broader HF patients in the general population.

The biggest limitation is that HFrEF etiology was reported by investigators and was not centrally adjudicated, and further investigation of "idiopathic" causes, including genetic analysis, might have identified a specific etiology. Misclassification of underlying cause cannot be excluded and could have attenuated the observed differences between groups. Although a Chagas etiology was not confirmed by serology, the geographic location of the cases (307 of 308 in Latin America), the typical baseline characteristics (eg, high proportion of women; low prevalence of typical comorbidities such as diabetes, hypertension and coronary heart disease; high prevalence of prior stroke and conventional pacemaker implantation), high event rates (remarkably similar to the recent PARACHUTE-HF trial in patients with confirmed Chagas disease), and elevated risk for stroke compared with other etiologies, collectively, strongly suggest that this cohort did have Chagasic HFrEF.

In addition, patients with uncorrected significant primary valvular disease were excluded from all 3 trials; thus, valvular heart disease is likely underrepresented in the present cohort. However, the clinical characteristics and outcomes were consistent with what we know about the specific etiologies analyzed.

The "other" category within nonischemic HF was not subclassified in detail, which may have led to heterogeneity within this group. Presently, nearly all cohorts of Chagasic HF have originated from Latin America, and future studies are needed to determine whether clinical outcomes differ among patients from regions outside Latin America.

Finally, given the number of comparisons performed, findings supported by modest *P* values should be interpreted with caution.

CONCLUSIONS

Patients with Chagasic HF had substantially worse clinical outcomes than those with other HF etiologies, including a risk for stroke. Patients with ischemic and valvular etiologies also had high event rates (including stroke among those with valvular disease), but participants with a viral etiology had low event rates. Given the expansion of Chagas disease beyond Latin America, our results underscore the need for improved recognition, prevention, and management of this distinctive and deadly type of HF.

DATA AVAILABILITY The original data are available through the specific application process offered by each sponsor of the trials included in the present analyses.

FUNDING SUPPORT AND AUTHOR DISCLOSURES

ATMOSPHERE and PARADIGM-HF were sponsored by Novartis, and GALACTIC-HF was sponsored by Amgen, Cytokinetics, and Servier. Dr Ono has received research grants from the Japanese College of Cardiology, the Kanzawa Medical Research Foundation, the Fukuda Foundation for Medical Technology, and the Nakatomi Foundation. Dr Docherty's employer, the University of Glasgow, has been remunerated by AstraZeneca for his work on clinical trials. Dr Docherty has received speaker fees from AstraZeneca, Boehringer Ingelheim, Pharmacosmos, and Translational Medicine Academy; has served on advisory boards or performed consultancy for Abbott Laboratories, FIRE-1, Us2.ai, and Bayer; has served on a clinical endpoint committee for Bayer; and has received research grant support (paid to his institution) from AstraZeneca, Roche Diagnostics, Novartis, and Boehringer Ingelheim. Dr Henderson has received personal fees from Bayer outside the submitted work. Dr Echeverria has received research support from Pfizer, Novartis, and Boehringer Ingelheim; and has received consulting fees from Novartis, Boehringer Ingelheim, AstraZeneca, and Bayer. Dr Felker has received grant funding to his institution from the American Heart Association, Amgen, Bayer, Bristol Myers Squibb, CSL Behring, Cytokinetics, Merck, MyoKardia, and the National Institutes of Health; has received consulting fees from Abbott Laboratories, American Regent, Amgen, AstraZeneca, Boehringer Ingelheim, Bristol Myers Squibb, Cardionomic, Cytokinetics, Medtronic, Myovant, Novartis, Reprieve, Sequana, Windtree Therapeutics, and WhiteSwell; and has participated on data and safety monitoring boards or advisory boards for EBR Systems, LivaNova, Medtronic, Siemens, Rocket Pharma, and V-Wave. Dr Lopes has received research grants from or has contracts with Bristol Myers Squibb, GlaxoSmithKline, Pfizer, Novartis, Ionis Pharmaceuticals, Regeneron Pharmaceuticals, Alnylam, Roche, Bayer, VeraDermics, Priovant Therapeutics, Basilea Pharmaceutica, Theravance Biopharma, and Cytokinetics; has received funding for educational activities or lectures from Pfizer, Novartis, Novo Nordisk, and Bristol Myers Squibb; and has received funding for consulting from Bayer, Boehringer Ingelheim, Bristol Myers Squibb, Novo Nordisk, Novartis, and Pfizer. Dr Martinez has received consulting fees and research grants from AstraZeneca, Baliarda, Bayer, Boehringer Ingelheim, Bristol Myers Squibb, Gador, Milestone, Novartis, Pfizer, and St Luke's University. Dr Ramires has received speaker fees from AstraZeneca, Amgen, Pfizer, Novartis, and Bristol Myers Squibb; and has received consulting fees from Novartis and Bristol Myers Squibb. Dr Packer has received consulting fees from 89bio, AbbVie, Altimmune, Amgen, Ardelyx, AstraZeneca, Boehringer Ingelheim, Caladrius, Casana, CSL Behring, Cytokinetics, Eli Lilly, Moderna, Novartis, Reata, Regeneron, Relypsa, and Salamandra; and is a trial executive committee member for the Boehringer Ingelheim and Eli Lilly & Company Diabetes Alliance. Dr Teerlink has received personal fees as chair of the GALACTIC-HF executive committee from Amgen and Cytokinetics; has received personal fees for research contracts from 3ive Labs, Amgen, AskBio, AstraZeneca, Bayer, Boehringer Ingelheim, Cardiac Dimensions, Cytokinetics, EBR Systems, Edgewise Therapeutics, Edwards

Lifesciences, Endotronix, Impulse Dynamics, LivaNova, Medtronic, Myovant, the Patient-Centered Outcomes Research Institute, Recardio, Tectonic, and V-Wave; is a consultant for 3ive Labs, AOP Medical, Arena, AstraZeneca, Bayer, Boehringer Ingelheim, Bristol Myers Squibb, Cardiac Dimensions, CorHepta, Cytokinetics, Daiichi-Sankyo, EBR Systems, Edwards Lifesciences, Impulse Dynamics, JuvLabs, Kaiser Permanente, Eli Lilly, LivaNova, Medtronic, Myovant, Novartis, the Patient-Centered Outcomes Research Institute, Pfizer, Recardio, ReCor Medical, Regeneron, Reprieve, Stealth, Tectonic, V-Wave, Verily, ViCardia, and Windtree Therapeutics; is a data monitoring committee member for Angitia, Cardurion, Kaiser Permanente, and Medtronic; and has served as secretary, treasurer, and president of Heart Failure Society of America. Dr Vaduganathan has received research grant support from, served on advisory boards for, or had speaker engagements with Alnylam Pharmaceuticals, American Regent, Amgen, AstraZeneca, Bayer, Baxter Healthcare, Bristol Myers Squibb, Boehringer Ingelheim, Chiesi, Cytokinetics, Esperion, Fresenius Medical Care, Idorsia Pharmaceuticals, Lexicon Pharmaceuticals, Merck, Milestone Pharmaceuticals, Novartis, Novo Nordisk, Pharmacosmos, Recordati, Relypsa, Roche Diagnostics, Sanofi, and Tricog Health; and participates on clinical trial committees for studies sponsored by Amgen, AstraZeneca, Boehringer Ingelheim, Galmed, Novartis, Bayer, Occlutech, Pharmacosmos, and Impulse Dynamics. Dr Zile has received research funding from Novartis; and has served as a consultant for Novartis, Abbott Laboratories, Boston Scientific, CVRx, EBR, Endotronix, Ironwood, Merck, Medtronic, and MyoKardia V Wave. Dr Jhund has received speaker fees from AstraZeneca and ProAdWise Communications; has received advisory board fees from AstraZeneca and Bayer; has received research funding from AstraZeneca, Boehringer Ingelheim, Analog Devices, and Roche Diagnostics; and is director of Global Clinical Trial Partners; Dr Jhund's employer, the University of Glasgow, has been remunerated for clinical trial work by AstraZeneca, Bayer, Novartis, and Novo Nordisk. Dr Solomon has received research grants from Actelion, Alnylam, Amgen, AstraZeneca, Bellerophon, Bayer, Bristol Myers Squibb, Celladon, Cytokinetics, Eidos, Gilead, GlaxoSmithKline, Ionis, Eli Lilly, Mesoblast, MyoKardia, the National Institutes of Health/National Heart, Lung, and Blood Institute, Neurotronik, Novartis, Novo Nordisk, Respicardia, Sanofi Pasteur, Theracos, and S2.AI; and has consulted for Abbott Laboratories, Action, Akros, Alnylam, Amgen, Arena, AstraZeneca, Bayer, Boehringer Ingelheim, Bristol Myers Squibb, Cardior, Cardurion, Corvia, Cytokinetics, Daiichi-Sankyo, GlaxoSmithKline, Eli Lilly, Merck, MyoKardia, Novartis, Roche, Theracos, Quantum Genomics, Cardurion, Janssen, Cardiac Dimensions, Tenaya, Sanofi Pasteur, Dinaqor, Tremeau, CellPro-Thera, Moderna, American Regent, and Sarepta. Dr McMurray has received payments to Glasgow University for clinical trials and other research projects from the British Heart Foundation, the National Institutes of Health/National Heart, Lung, and Blood Institute, Alynylam Pharmaceuticals, AstraZeneca, Bayer, Cardurion, Cytokinetics, Novartis, and Roche Diagnostics; has received personal consultancy fees from Alynylam Pharmaceuticals, AnaCardio, AstraZeneca, Bayer, Cardurion, Cytokinetics, Novartis, River BioMedics, Biohaven Pharmaceuticals, and DalCor Pharmaceuticals; has received personal lecture fees from Alkem Metabolics, AstraZeneca, Canadian Medical and Surgical Knowledge, Centrix Healthcare, Emcure Pharmaceuticals, Eris Lifesciences, Hikma Pharmaceuticals, Imagica Health, Intas Pharmaceuticals, J.B. Chemicals & Pharmaceuticals, Lupin Pharmaceuticals, Medscape/Heart.org, ProAdWise Communications, Radcliffe Cardiology, Sun Pharmaceuticals, Translational Medicine Academy, Regeneron, MCI India, Hilton Pharmaceuticals, IMEDIC Pharmaceuticals Micro Labs, At the Limits, and ARMGO Pharmaceuticals; is a data and safety monitoring board member for WCG Clinical Services; and is a director of Global Clinical Trial Partners (which provides clinical trial services such as endpoint committees and educational programs).

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KEY WORDS Chagas, clinical outcomes, etiology, heart failure with reduced ejection fraction, HFrEF, nonischemic

APPENDIX For supplemental tables, please see the online version of this paper.